

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.363243086	delta=+0.0116455	
ratio=1.0049522			
CoG x [m]	-0.002024708562 -> -0.03381353344	delta=-0.0317888	
ratio=-			
CoG y [m]	-3.052657894e-05 -> -0.0372118264	delta=-0.0371813	
ratio=-			
CoG z [m]	0.009509749599 -> 0.01162284223	delta=+0.00211309	
ratio=-			
rotor effectiveness 1	1 -> 1.062483085	delta=+0.0624831	ratio=1.0624831
rotor effectiveness 2	1 -> 0.9704751033	delta=-0.0295249	ratio=0.9704751
rotor effectiveness 3	1 -> 0.8501317769	delta=-0.149868	ratio=0.85013178
rotor effectiveness 4	1 -> 0.9322984059	delta=-0.0677016	ratio=0.93229841
command lag [s]	0 -> 0.004	delta=+0.004	ratio=-

Inertia matrix [kg m²]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06599724502, -0.0008710728461, -0.0008579459497]
[-0.0008710728461, 0.06558598533, 0.0003813693409]
[-0.0008579459497, 0.0003813693409, 0.1289746596]
```

Principal moments: reference -> estimated

principal moment 1 [kg m ²]	0.06495264516 -> 0.06489574419	delta=-5.6901e-05	
ratio=0.99912396			
principal moment 2 [kg m ²]	0.06500006701 -> 0.06667336157	delta=+0.00167329	
ratio=1.025743			
principal moment 3 [kg m ²]	0.1289921163 -> 0.1289887842	delta=-3.33215e-06	
ratio=0.99997417			

Optimization

joint dynamics loss: 569.653647038
Gaussian-prior cost: 13.666886821
selected strict-ZOH lag [s]: 0.004

Bag failure_2_single

reconstruction position RMSE [m]: 66.3248755038
reconstruction position component RMSE [m]: [11.603111 3.696807 65.19732]
reconstruction orientation RMSE [deg]: 147.783693338
gyro consistency RMSE xyz [rad/s]: [17.458329 17.086403 23.260544]
specific-force consistency RMSE xyz [m/s²]: [51.354815 61.691816 33.760009]
inferred external force RMS xyz [N]: [7.123979 14.328369 5.210622]
inferred external torque RMS xyz [N m]: [2.865367 3.585207 8.272134]
bag dynamics loss: 569.653647038

Recorded command timestamp intervals (data-derived)

Bag failure_2_single

rotor: median dt=0.00495767593 s (201.707 Hz), min=2.14576721e-06 s, max=0.15852499 s
gimbal: median dt=0.0049571991 s (201.727 Hz), min=2.62260437e-06 s, max=0.158331871 s